

JONQUIERE, QC, Canada

WMO: 716170

Lat: 48.4250N

Lon: 71.1425W

Elev: 136

StdP: 99.71

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.2	-26.5	-33.1	0.2	-29.0	-30.4	0.2	-26.2	12.1	-9.7	11.2	-12.7	3.0	280	0.642

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	29.0	19.7	27.1	19.0	25.3	18.3	21.7	26.2	20.7	24.8	19.6	23.6	4.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.2	15.2	24.0	19.2	14.2	22.7	18.1	13.3	21.8	64.1	26.4	60.3	24.9	56.7	23.5	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	9.3	8.3	DB	-32.8	32.2	3.2	1.7	-35.1	33.5	-37.0	34.5	-38.8	35.4	-41.2	36.7	(4)
(5)			WB	-33.1	23.5	3.2	1.2	-35.4	24.4	-37.2	25.1	-39.0	25.8	-41.4	26.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.7	-14.3	-12.2	-5.7	2.5	10.7	16.1	18.8	17.6	13.1	6.4	-0.9	-8.9	(6)	
	DBStd	12.57	7.67	6.99	6.58	4.71	4.39	3.87	2.84	3.06	4.15	4.14	5.18	6.61	(7)	
	HDD10.0	3197	754	622	488	228	44	2	0	0	15	128	328	587	(8)	
	HDD18.3	5453	1013	856	746	476	241	88	28	50	163	371	577	846	(9)	
	CDD10.0	888	0	0	1	2	65	186	273	237	109	16	1	0	(10)	
	CDD18.3	103	0	0	0	0	3	22	43	28	7	0	0	0	(11)	
	CDH23.3	887	0	0	0	1	67	233	321	206	57	2	0	0	(12)	
	CDH26.7	201	0	0	0	0	17	63	68	42	10	0	0	0	(13)	
(14) Wind	WSAvg	3.8	4.1	4.1	4.2	4.3	3.9	3.6	3.2	3.2	3.3	3.9	4.2	3.9	(14)	
(15) Precipitation	PrecAvg	949	52	48	51	57	77	100	113	100	101	92	67	76	(15)	
	PrecMax	1069	86	94	106	93	162	154	184	224	203	146	127	108	(16)	
	PrecMin	834	15	21	25	25	30	39	42	38	54	45	19	39	(17)	
	PrecStd	76	19	18	22	23	30	28	36	44	32	29	21	18	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.6	5.9	11.3	19.7	29.0	31.5	31.0	30.2	28.0	21.7	14.8	5.6	(19)	
		MCWB	4.6	4.5	5.8	10.6	16.3	21.0	21.1	20.8	20.7	15.5	12.0	4.3	(20)	
	2%	DB	2.4	2.1	7.0	15.2	25.0	28.4	28.6	27.7	24.8	18.0	10.3	3.0	(21)	
		MCWB	1.7	0.7	3.3	7.7	14.4	18.8	20.0	20.0	19.3	13.4	8.1	1.9	(22)	
	5%	DB	-0.3	0.4	4.7	12.1	21.8	26.0	26.7	25.6	22.2	14.9	7.6	1.4	(23)	
		MCWB	-1.1	-0.7	1.9	6.2	13.3	17.8	19.3	18.9	17.2	11.2	5.8	0.6	(24)	
	10%	DB	-3.2	-1.4	3.0	9.4	19.0	23.7	25.0	23.8	20.0	12.5	5.6	0.1	(25)	
		MCWB	-4.0	-2.4	1.0	5.0	12.2	16.7	18.7	17.9	16.1	9.8	4.1	-0.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.6	4.5	7.1	12.2	18.9	22.7	23.4	22.4	21.8	16.3	12.1	4.4	(27)	
		MCDB	5.3	5.7	9.7	18.1	24.0	28.8	27.8	27.3	26.6	20.6	14.7	5.3	(28)	
	2%	WB	1.7	0.9	3.8	8.6	16.6	20.9	21.7	21.1	19.9	13.9	8.5	2.0	(29)	
		MCDB	2.5	1.8	5.9	13.1	22.2	25.6	25.7	25.6	23.4	17.2	9.9	2.7	(30)	
	5%	WB	-1.1	-0.7	2.4	7.0	14.4	19.4	20.8	20.0	18.1	11.8	6.0	0.7	(31)	
		MCDB	-0.3	0.3	4.2	11.2	19.5	23.5	24.6	24.2	21.5	14.1	7.3	1.4	(32)	
	10%	WB	-4.0	-2.6	1.1	5.5	12.7	17.9	19.6	19.0	16.6	10.0	4.2	-0.5	(33)	
		MCDB	-3.2	-1.5	2.7	8.8	18.2	22.2	23.5	22.6	19.5	12.1	5.5	0.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	9.9	9.8	9.4	11.4	10.9	10.2	10.1	9.6	7.3	6.2	7.2	(35)	
		MCDBR	10.7	9.2	10.8	14.5	16.5	15.2	13.2	13.1	12.9	11.7	9.1	7.4	(36)	
	5% WB	MCWBR	10.4	8.4	7.8	8.7	8.1	7.6	6.3	6.6	7.6	7.7	7.5	7.0	(37)	
		MCWBR	10.6	9.1	9.1	12.6	13.4	12.0	10.9	10.8	11.3	10.4	8.4	7.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.266	0.285	0.315	0.378	0.359	0.385	0.403	0.382	0.370	0.329	0.317	0.265	(40)	
		taud	2.259	2.218	2.204	2.220	2.405	2.360	2.324	2.383	2.403	2.502	2.369	2.318	(41)	
	Ebn at Noon		801	868	896	869	903	877	854	857	832	818	732	756	(42)	
		Edh at Noon	74	98	118	130	112	119	121	109	97	75	68	62	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.31	2.27	3.52	4.46	5.16	5.50	5.19	4.41	3.38	1.93	1.05	0.86	(44)	
	RadStd		0.16	0.30	0.29	0.35	0.49	0.37	0.51	0.29	0.25	0.20	0.11	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page